YEAR 7–8 MATHEMATICS

Angles and Measurement

Practice Questions

Work in the space under each question. Give a reason for every angle fact. Answers are in the separate Answers booklet.

SECTION	CORE QUESTIONS ★	ALL	CORE
Part A Angles			
A1 Points, lines, angles and notation	1, 4, 5	12 min	8 min
A2 Measuring and classifying angles	2, 4, 5	11 min	8 min
A3 Angles at a point	1, 2, 4	20 min	12 min
A4 Transversals and parallel lines	1, 2, 4	18 min	11 min
A5 Further parallel-line problems and proving lines parallel	1, 2, 5	19 min	13 min
A6 Triangles	1, 2, 5	17 min	10 min
A7 Quadrilaterals	1, 3, 5	16 min	12 min
Part B Measurement			
B1 Units and conversions	1, 4, 6	13 min	7 min
B2 Perimeter	1, 3, 4	14 min	8 min
B3 Area of rectangles and composite shapes	1, 4, 5	14 min	10 min
B4 Areas of triangles and parallelograms	1, 4, 5	13 min	9 min
B5 Special quadrilaterals: rhombus, kite, trapezium	1, 5, 6	13 min	8 min
Part C Mixed challenge			
C Review and challenge questions	3	29 min	3 min
Total		3 h 29 min	1 h 59 min

This pack holds more than one lesson's worth of questions, so choose questions to suit the student. The questions marked ★ make a suggested core set for a 2-hour lesson: one warm-up in each section, then harder questions. Every question is taken word for word from ICE-EM Mathematics 7 (3rd edition) or Cambridge Maths NSW Year 8 (2nd edition), and its source is shown beside it. Times are a guide for a typical Year 7 or Year 8 student.

PART A

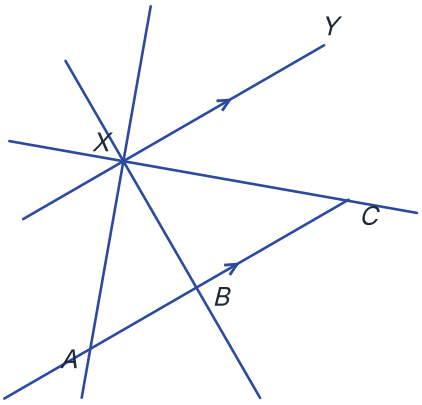
Angles

Points, lines and angle notation; measuring and classifying angles; angles at a point; transversals and parallel lines; proving lines parallel; triangles; quadrilaterals.

A1 POINTS, LINES, ANGLES AND NOTATION · Y7 ICE-EM 6A–6B · Y8 Cambridge 6A

About 12 minutes. Core: questions 1, 4 and 5.

1 ★ Look at the diagram. Y7 ICE-EM Ex 6A Q3



a Name the four concurrent lines passing through the point X.

b Name the two parallel lines, using the correct symbol for 'is parallel to'.

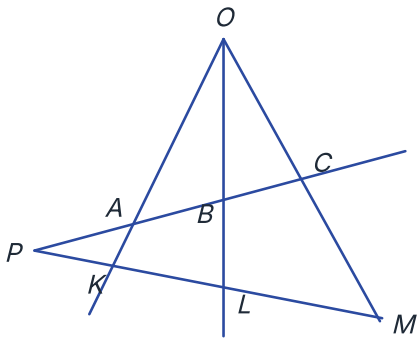
c Name the intersection of the lines BX and AC.

d Name the line joining C and the intersection of AX and BX.

e What points belong to both lines AC and CB?

f What points belong to both lines AC and XY?

2 There are five lines in the diagram. Y7 ICE-EM Ex 6B Q2



a Copy and complete: The lines AK, BL and CM are _____.

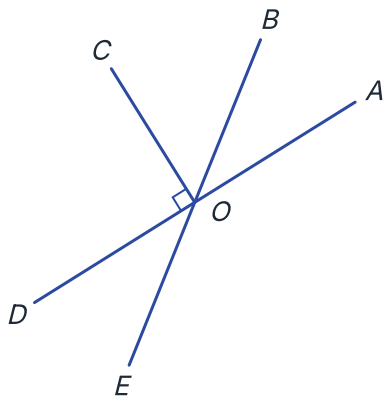
b Does L lie on the ray OB?

c Does L lie on the interval PK?

d Name the two rays with vertex P.

e Name the four rays with vertex A.

3 Consider the given diagram. Y8 Cambridge Ex 6A Q5

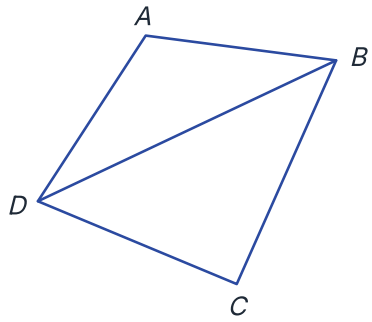


a Name an angle that is:
i vertically opposite to $\angle AOB$
ii complementary to $\angle BOC$
iii supplementary to $\angle AOE$
iv supplementary to $\angle AOC$

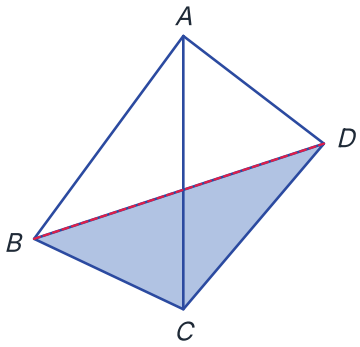
b Copy and complete:
i $AD \perp$ ___
ii ___ $\perp OD$

4 ★ Y7 ICE-EM Ex 6B Q7

a Name all eight angles associated with the figure below.

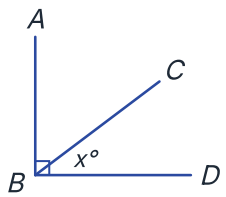


b Name all 12 angles formed at the vertices of the tetrahedron below.

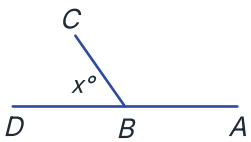


5 ★ Write down an algebraic expression for the size of the $\angle ABC$. Y8 Cambridge Ex 6A Q10

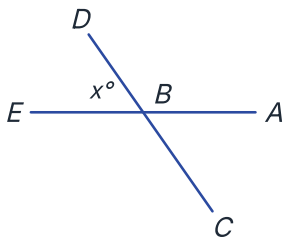
a



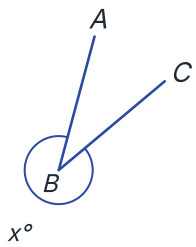
b



c



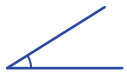
d



A2 MEASURING AND CLASSIFYING ANGLES · Y7 ICE-EM 6C

About 11 minutes. Core: questions 2, 4 and 5.

1 Classify each angle as acute, obtuse, reflex or right. Y7 ICE-EM Ex 6C Q3

a 

b 

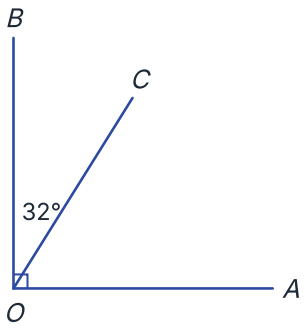
c 

d 

2 ★ Write down the complement of: Y7 ICE-EM Ex 6C Q1

a 15° b 35° c 72° d 88° e 56°

3 Find $\angle AOC$ in the diagram. Also give the size of reflex angle $\angle AOC$. Y7 Ch 6 Review Q4



4 ★ What is the size of the angle between the hour hand and the minute hand at: Y7 ICE-EM Ex 6C Q4

a 6 a.m.? b 3 a.m.? c 1 p.m.? d 2 p.m.?

e 4 a.m.? f 8 p.m.? g 10 a.m.? h 7 a.m.?

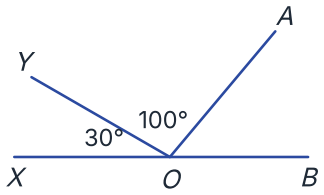
5 ★ How many minutes pass after 12 noon before the angle between the minute and hour hand is exactly 90°? Y7 Ch 6 Challenge Q7

A3 ANGLES AT A POINT · Y7 ICE-EM 6D · Y8 Cambridge 6A

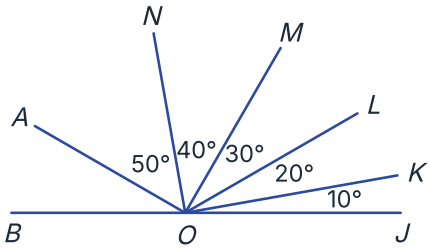
About 20 minutes. Core: questions 1, 2 and 4.

1 ★ Find $\angle AOB$ in each diagram below, giving reasons for your answer. Y7 ICE-EM Ex 6D Q5

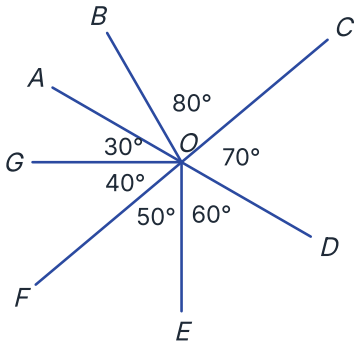
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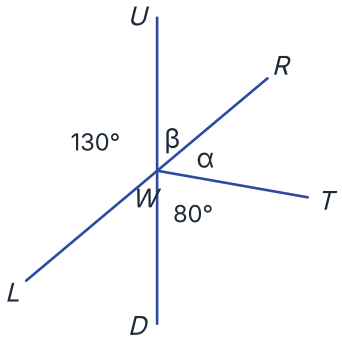


l

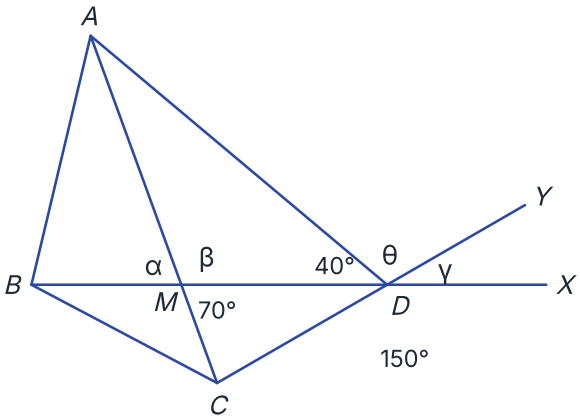


2 ★ Find the values of α, β, γ and θ in each diagram below, carefully giving reasons for all of your statements. Y7 ICE-EM Ex 6D Q7

g



h



3 A round birthday cake is cut into sectors for nine friends (including Jack) at Jack's birthday party. After the cake is cut, there is no cake remaining. What will be the angle at the centre of the cake for Jack's piece if: Y8 Cambridge Ex 6A Q8

a everyone receives an equal share?

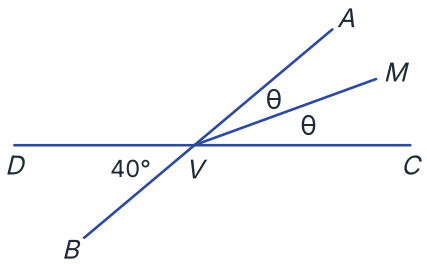
b Jack receives twice as much as everyone else? (In parts **b**, **c** and **d** assume his friends have an equal share of the rest.)

c Jack receives four times as much as everyone else?

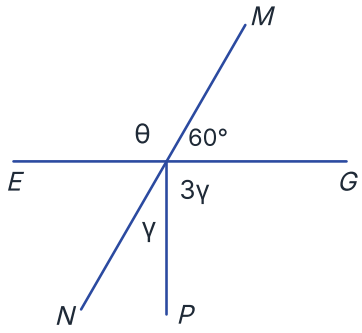
d Jack receives ten times as much as everyone else?

4 ★ In each diagram below, angles marked with the same Greek letter are equal in size. Find the value of each, giving reasons for your answers. Y7 ICE-EM Ex 6D Q8

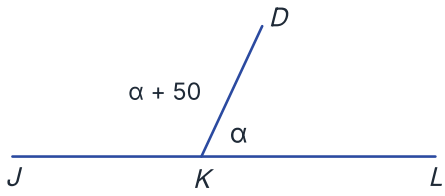
c



g



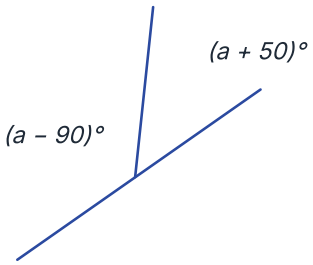
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5 Consider this diagram (not drawn to scale). Y8 Cambridge Ex 6A Q13

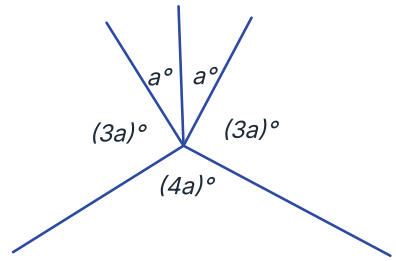
a Calculate the value of a.

b Explain what is wrong with the way the diagram is drawn.

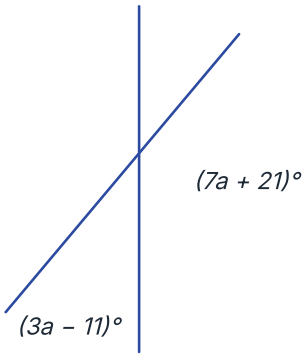


6 Equations can be helpful in solving geometric problems in which more complex expressions are involved. Find the value of a in these diagrams. Y8 Cambridge Ex 6A Q14

e



f



A4 TRANSVERSALS AND PARALLEL LINES · Y7 ICE-EM 6E · Y8 Cambridge 6B

About 18 minutes. Core: questions 1, 2 and 4.

1 ★ Name the angle that is: Y8 Cambridge Ex 6B Q2

a corresponding to $\angle ABF$

b corresponding to $\angle BCG$

c alternate to $\angle FBC$

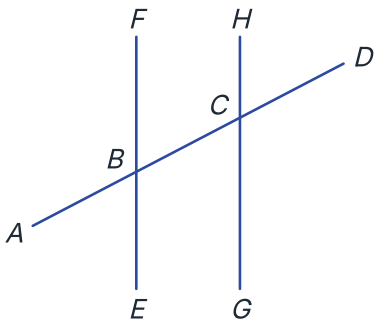
d alternate to $\angle CBE$

e cointerior to $\angle HCB$

f cointerior to $\angle EBC$

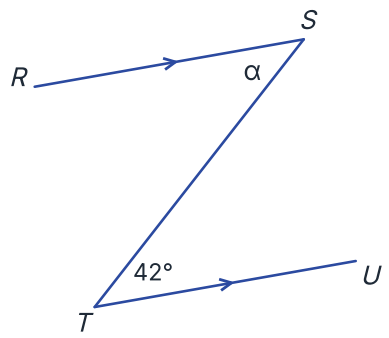
g vertically opposite to $\angle ABE$

h vertically opposite to $\angle HCB$

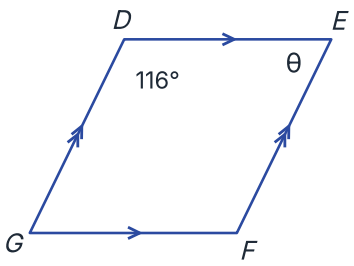


2 ★ Find the values of α, β, γ and θ in the diagrams below. Give careful reasons for all your statements, mentioning the relevant parallel lines. Y7 ICE-EM Ex 6E Q7

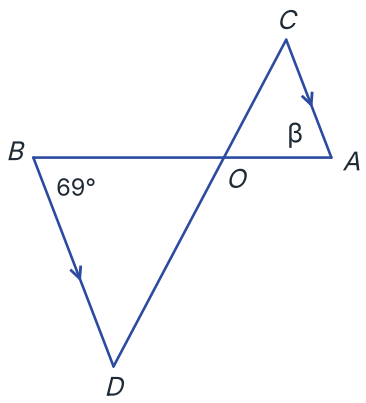
d



g

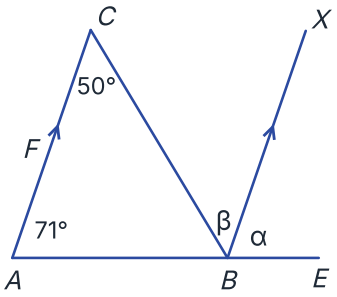


k

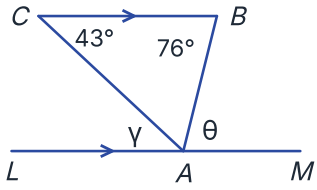


3 Find the values of the letters α, β, γ and θ in the diagrams below. Give a reason for each step in your argument, and name the relevant parallel lines. Y7 ICE-EM Ex 6E Q8

f

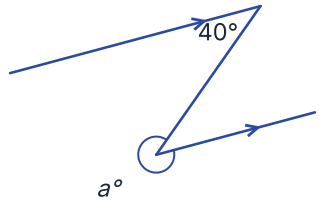


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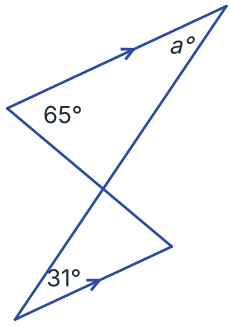


4 ★ Find the value of a in these diagrams. Y8 Cambridge Ex 6B Q7

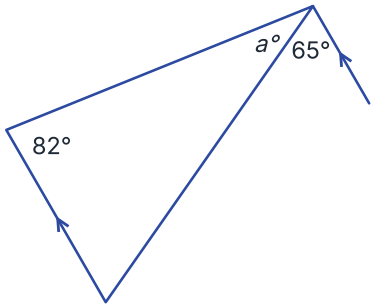
b



e

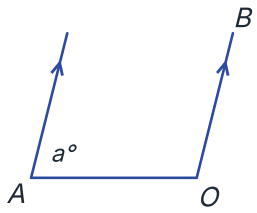


i

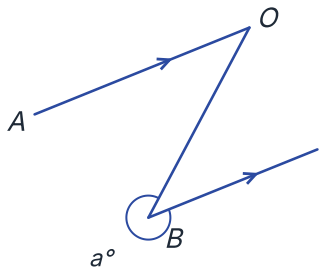


5 Write an algebraic expression for the size of $\angle AOB$. Y8 Cambridge Ex 6B Q9

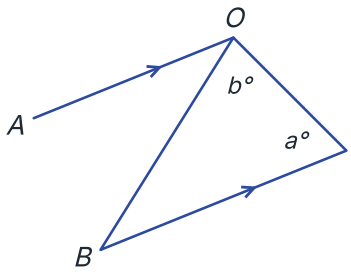
a



b



c

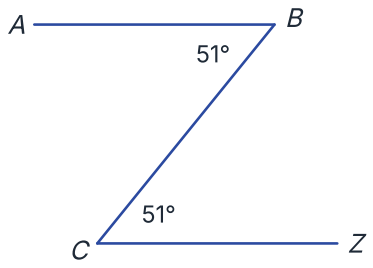


A5 FURTHER PROBLEMS AND PROVING LINES PARALLEL · Y7 ICE-EM 6F–6G

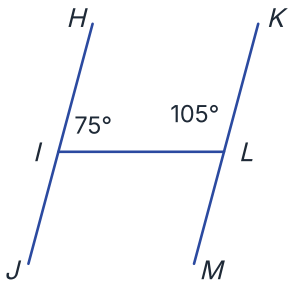
About 19 minutes. Core: questions 1, 2 and 5.

1 ★ For each diagram below, name all pairs of parallel lines, giving reasons. Y7 ICE-EM Ex 6G Q1

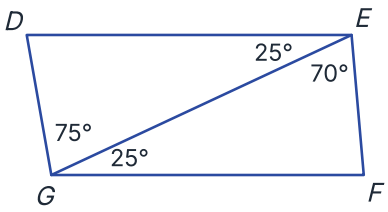
a



c

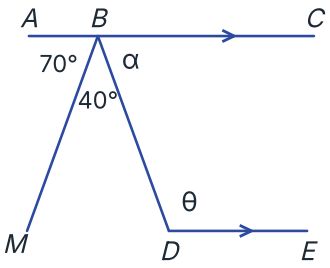


f

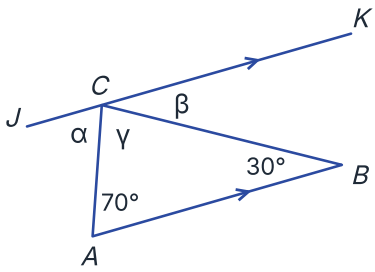


2 ★ Find the values of α, β, γ and θ in each of the diagrams below. Give careful reasons for all your statements. Y7 ICE-EM Ex 6F Q1

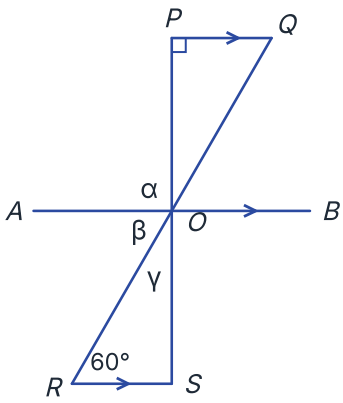
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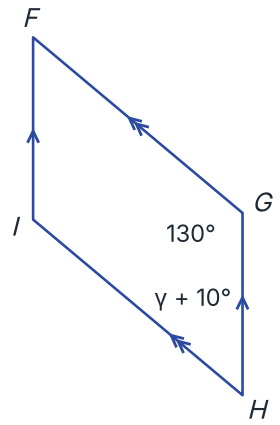


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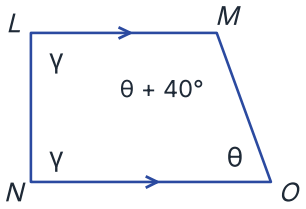


3 Find the values of α, β, γ and θ, giving reasons for your answers. Y7 ICE-EM Ex 6F Q4

d

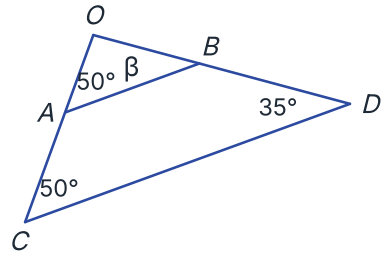


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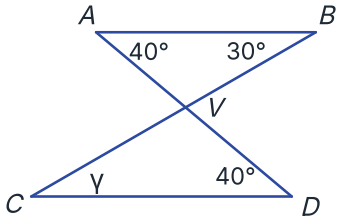


4 In each diagram below, give a reason why $AB \parallel CD$. Hence, find the values of α, β, γ and θ. Give all reasons. Y7 ICE-EM Ex 6G Q2

b

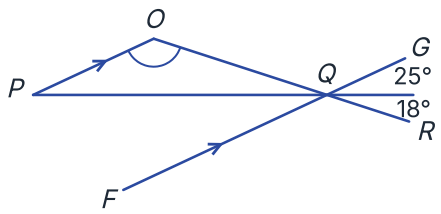


d

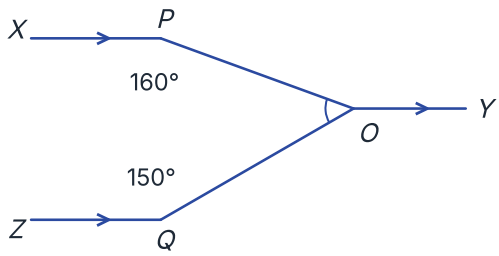


5 ★ Find the size of the marked angle $\angle POQ$ in each diagram below. Each solution will require two or more steps, each with its own reason. Y7 ICE-EM Ex 6F Q2

b



d



A6 TRIANGLES · Y8 Cambridge 6C

About 17 minutes. Core: questions 1, 2 and 5.

1 ★ Give the common name of a triangle with these properties. Y8 Cambridge Ex 6C Q1

a one right angle

b two equal side lengths

c all angles acute

d all angles 60°

e one obtuse angle

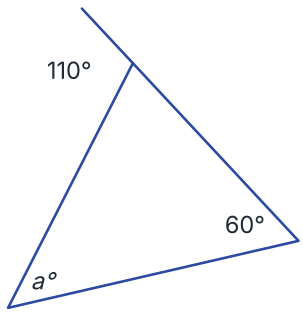
f three equal side lengths

g two equal angles

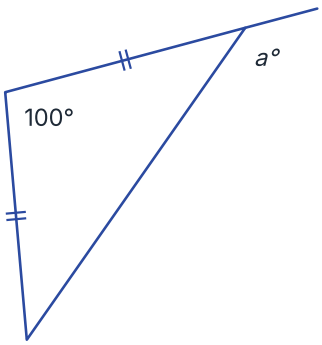
h three different side lengths

2 ★ Find the value of a. Y8 Cambridge Ex 6C Q7

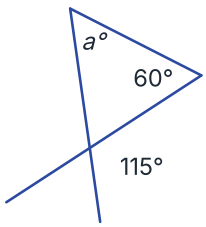
d



e



f



3 A triangle is constructed using two radii and a chord. Y8 Cambridge Ex 6C Q11

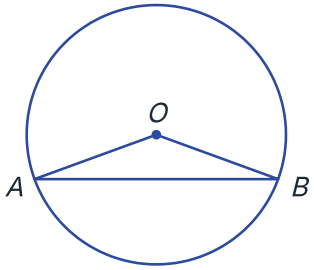
a What type of triangle is $\triangle AOB$? Explain.

b Name two angles that are equal.

c Find $\angle ABO$ if $\angle BAO$ is 30°.

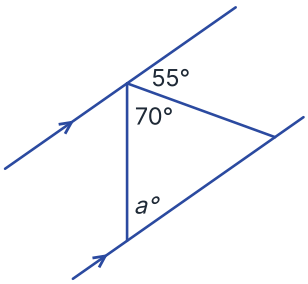
d Find $\angle AOB$ if $\angle OAB$ is 36°.

e Find $\angle ABO$ if $\angle AOB$ is 100°.

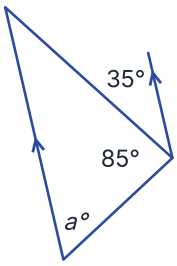


4 Use your knowledge of parallel lines and triangles to find the unknown angle a. Y8 Cambridge Ex 6C Q9

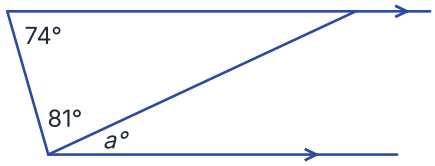
a



b

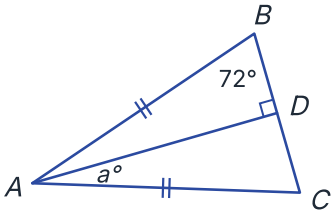


c

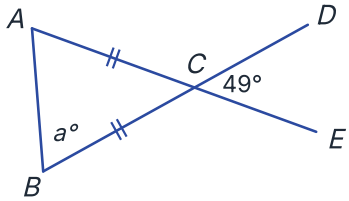


5 ★ Find the value of a in these diagrams, giving reasons. Y8 Cambridge Ex 6C Q10

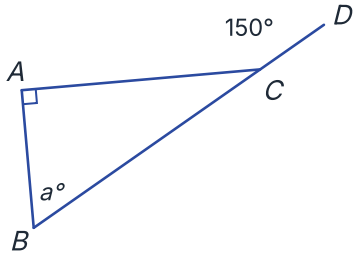
c



e



f

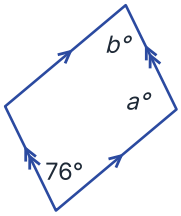


A7 QUADRILATERALS · Y8 Cambridge 6D

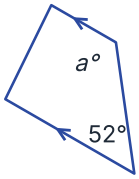
About 16 minutes. Core: questions 1, 3 and 5.

1 ★ Find the value of the pronumerals in these quadrilaterals. Y8 Cambridge Ex 6D Q3

a



c



2 By considering the properties of special quadrilaterals, decide if the following are always true. Y8 Cambridge Ex 6D Q9

a A square is a type of rectangle.

b A rectangle is a type of square.

c A square is a type of rhombus.

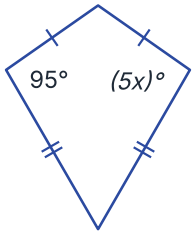
d A rectangle is a type of parallelogram.

e A parallelogram is a type of square.

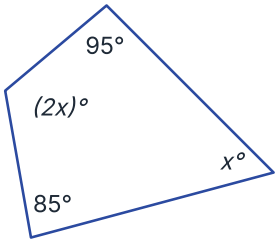
f A rhombus is a type of parallelogram.

3 ★ Use your algebra skills to find the value of x in these diagrams. Y8 Cambridge Ex 6D Q7

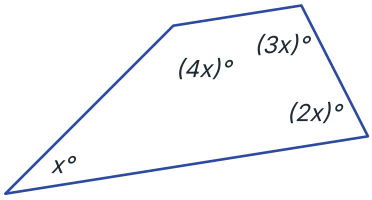
a



b



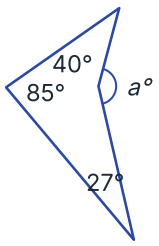
c



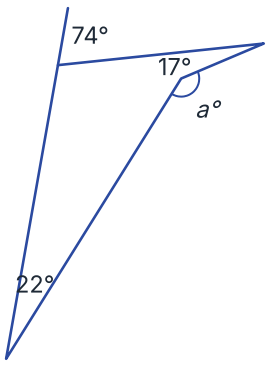
4 Is it possible to draw a non-convex quadrilateral with two or more interior reflex angles? Explain and illustrate. Y8 Cambridge Ex 6D Q10

5 ★ Use your knowledge of geometry from the previous sections to find the values of a. Y8 Cambridge Ex 6D Q6

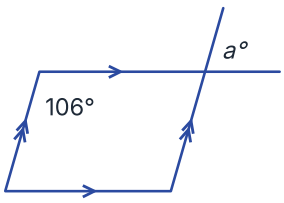
a



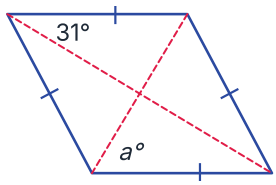
c



d



e



PART B

Measurement

Units and conversions; perimeter; areas of rectangles and composite shapes; areas of triangles and parallelograms; areas of the rhombus, kite and trapezium.

B1 UNITS AND CONVERSIONS · Y7 ICE-EM 9A–9B

About 13 minutes. Core: questions 1, 4 and 6.

1 ★ Copy and complete these statements. Y7 ICE-EM Ex 9A Q2

e 27 mm = ____ cm **h** 1.9 m = ____ cm **k** 6.34 km = ____ m

l 200 000 mm = ____ km **n** 2901 m = ___ km **s** $\frac{5}{8}$ cm = ____ mm

2 If you are 158 cm tall, how many decimetres is that? Y7 ICE-EM Ex 9B Q6

3 Markers on the road are placed 20 m apart. If the distance from the first marker to the last one is 1 km, how many markers are there in total? Y7 ICE-EM Ex 9A Q9

4 ★ Nine bus stops are equally spaced along a road. The distance from the first to the third stop is 600 m. How far is it from the first stop to the last? Y7 Ch 9 Challenge Q11

5 A man plants 12 rows of beans in rows that are 10 m long. The plants in each row are spaced 50 cm apart. If the first and last plants in a row are 25 cm from the ends of the row, what is the total number of plants? Y7 Ch 9 Challenge Q10

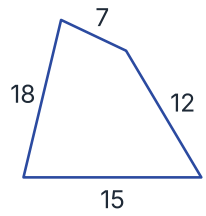
6 ★ In bygone days, the imperial system of measurement was used. In this system, distance is measured in inches, feet, yards and miles. There are 12 inches in a foot, 3 feet in a yard, and 1760 yards in a mile. Given that 2.54 cm equals 1 inch, how many kilometres is a mile? Y7 Ch 9 Review Q22

B2 PERIMETER · Y7 ICE-EM 9D

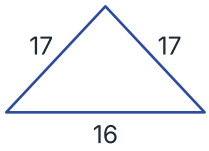
About 14 minutes. Core: questions 1, 3 and 4.

1 ★ Find the perimeter of each of these figures. Sides that are equal are marked with the same symbol. Y7 ICE-EM Ex 9D Q1

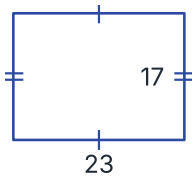
d



e

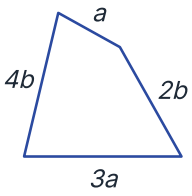


f

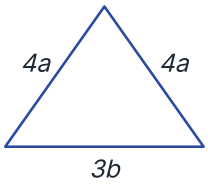


2 Write an algebraic expression for the perimeter of each figure. Y7 ICE-EM Ex 9D Q5

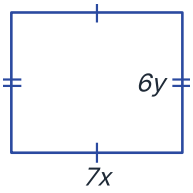
d



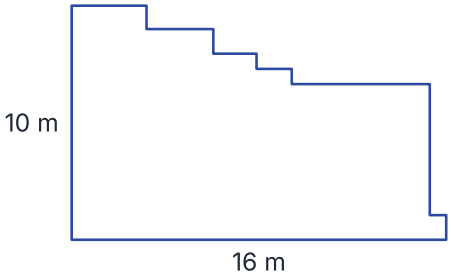
e



f



3 ★ Find the perimeter of the shape below. Once you figure out the trick, look at some of the previous perimeter exercises and see how quickly you can do them! Y7 Ch 9 Challenge Q14



4 ★ Investigate the possible perimeters for a rectangle of area 64 m^2. Which of the alternatives has the smallest perimeter? Y7 Ch 9 Challenge Q18

5 Chase and Lana have 108 m of fencing with which to enclose an area of the schoolyard for a rectangular garden. Explore the possible areas of their garden, and state which alternative has the largest area. Y7 Ch 9 Challenge Q19

B3 AREA OF RECTANGLES AND COMPOSITE SHAPES · Y7 ICE-EM 9E–9F

About 14 minutes. Core: questions 1, 4 and 5.

1 ★ Y7 ICE-EM Ex 9E Q4

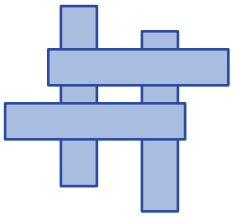
a Find the area in square metres of a rectangle with length 5 m and width 85 cm.

b Find the area in square kilometres of a rectangular region of land with length 800 m and width 2 km.

2 The table below gives incomplete information for four rectangles. Copy the table and fill in the gaps. Y7 ICE-EM Ex 9E Q7

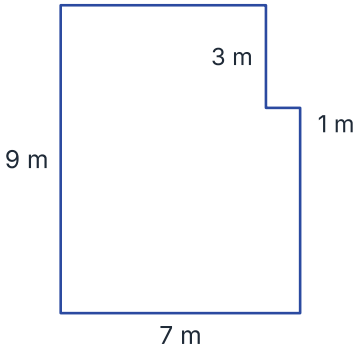
	Length	Width	Area
a	7 cm	6 cm	
b	9 cm		27 cm^2
c		17 m	85 m^2
d	12 km		144 km^2

3 Four rectangles, each with length 5 m and width 1 m, overlap to form the figure shown. Calculate its area. Y7 ICE-EM Ex 9F Q5

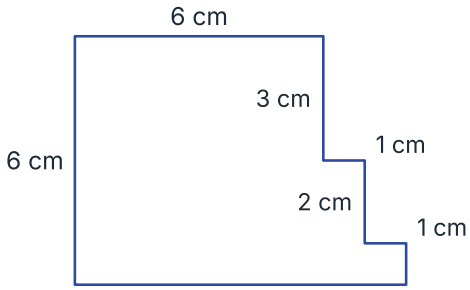


4 ★ Calculate the area of each figure. Y7 ICE-EM Ex 9F Q2

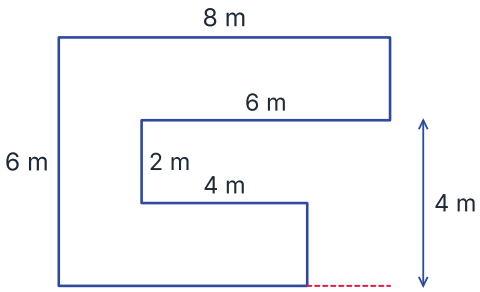
d



f



h



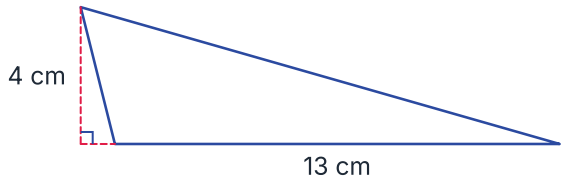
5 ★ The floors of three rooms of a palace are to be covered in tiles. Each tile is one centimetre square and costs $4.50. The rooms measure 8 m by 5.4 m, 2.7 m by 8.9 m and 300 cm by 13.3 m, respectively. Calculate the total cost. Y7 ICE-EM Ex 9E Q15

B4 AREAS OF TRIANGLES AND PARALLELOGRAMS · Y7 ICE-EM 9G

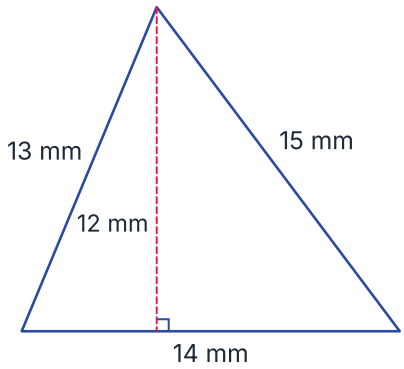
About 13 minutes. Core: questions 1, 4 and 5.

1 ★ Find the area of each triangle. Y7 ICE-EM Ex 9G Q1

c

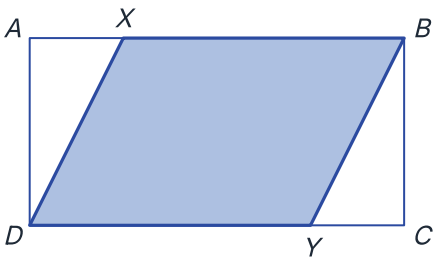


d



2 A parallelogram has a height that is half of its base. The area of the parallelogram is 18 cm^2. What is the height and base of the parallelogram? Y7 ICE-EM Ex 9G Q14

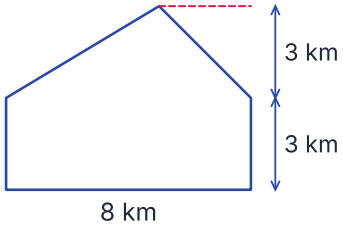
3 Rectangle $ABCD$ has length 8 cm and height 4 cm. Y7 ICE-EM Ex 9G Q15



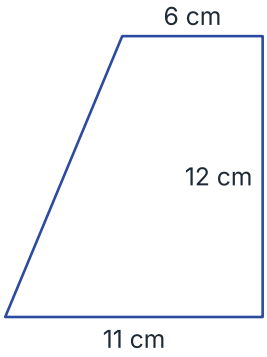
The two triangles AXD and BYC have height 4 cm and base 2 cm. Find the area of the parallelogram $DXBY$.

4 ★ Find the area of each figure below by adding the areas of squares, rectangles and triangles as necessary. Y7 ICE-EM Ex 9G Q7

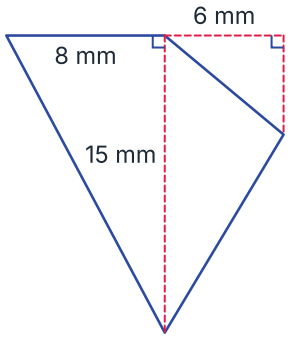
b



c

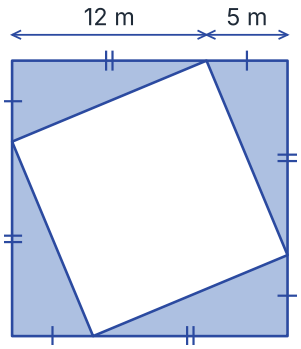


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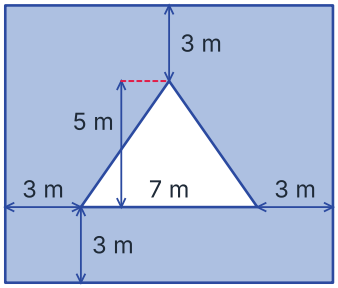


5 ★ Find the area of the shaded region in each figure below by adding and subtracting the areas of squares, rectangles and triangles as necessary. Y7 ICE-EM Ex 9G Q8

a



d

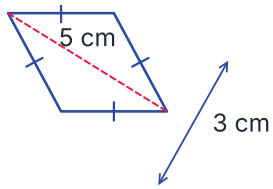


B5 SPECIAL QUADRILATERALS · Y8 Cambridge 3D

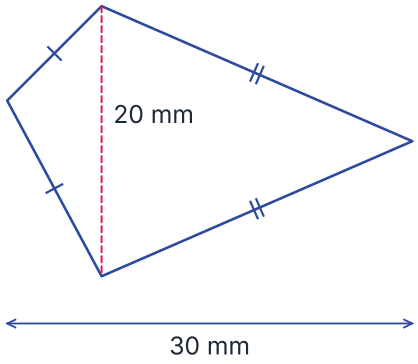
About 13 minutes. Core: questions 1, 5 and 6.

1 ★ Find the area of these special quadrilaterals. First give the name of the shape. Y8 Cambridge Ex 3D Q3

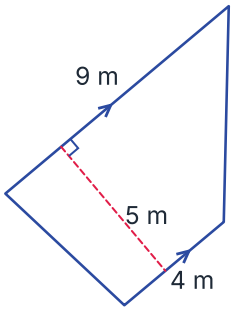
d



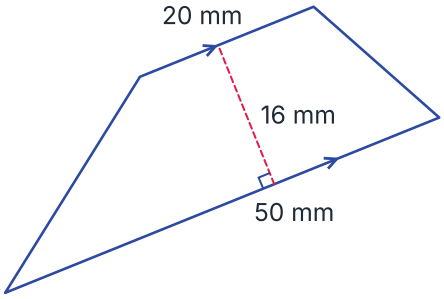
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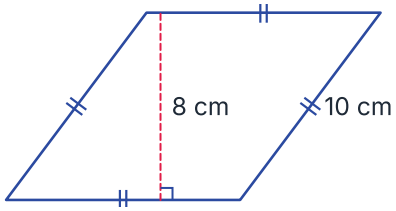


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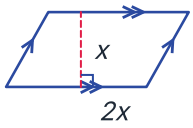
2 A landscape gardener charges $\$20$ per square metre of lawn. A lawn area is in the shape of a rhombus and its diagonals are 8 m and 14.5 m. What would be the cost of laying this lawn? Y8 Cambridge Ex 3D Q7

3 Would you use the formula $A = \frac{1}{2}xy$ to find the area of this rhombus? Explain. Y8 Cambridge Ex 3D Q11

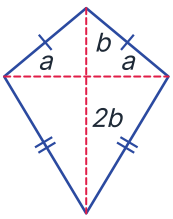


4 Write an expression for the area of these shapes in simplest form (e.g. $A = 2a + 3ab$). Y8 Cambridge Ex 3D Q10

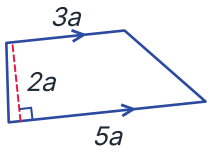
a



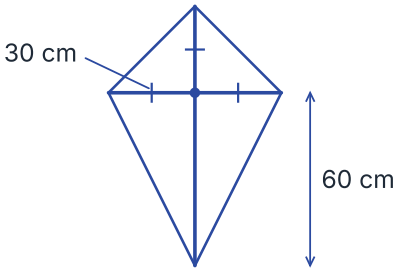
b



c



5 ★ A flying kite is made from four centre rods all connected near the middle of the kite, as shown. What area of plastic, in square metres, is needed to cover the kite? Y8 Cambridge Ex 3D Q5



6 ★ The parallel sides of a trapezium are 2 cm apart and one of the sides is three times the length of the other. If the area of the trapezium is 12 cm^2, what are the lengths of the parallel sides? Y8 Cambridge Ex 3D Q8

PART C

Mixed challenge

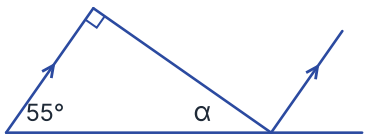
Review and challenge questions from the two Year 7 chapters: angles, then length, perimeter and area.

C MIXED CHALLENGE · Y7 ICE-EM Ch 6 and Ch 9 Review and Challenge

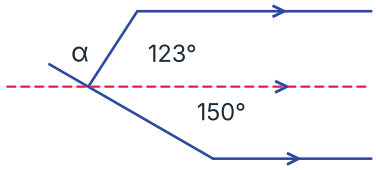
About 29 minutes. Core: question 3.

1 Find the value of α in each diagram below. Y7 Ch 6 Review Q7

b



d



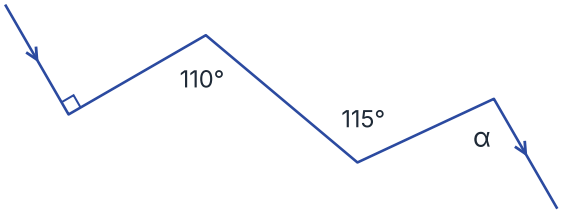
2 Triangle ABC has side lengths $AC = 4$ cm, $AB = 3$ cm and $BC = 5$ cm. $\angle BAC = 90°$. The point X is on BC and AX is perpendicular to BC. Y7 Ch 9 Review Q26

Find:

a the area of triangle ABC

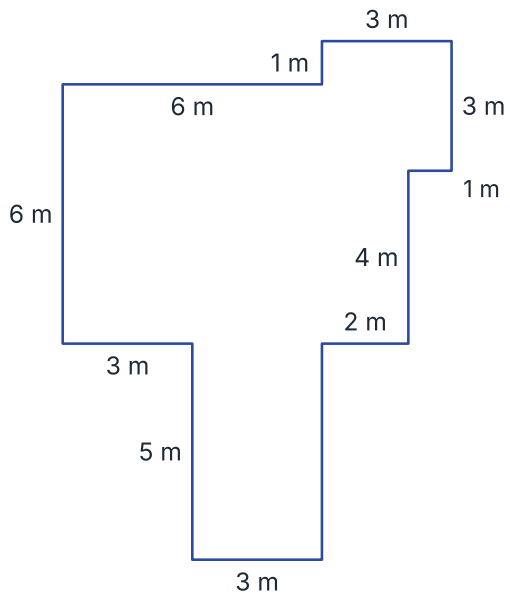
b the length AX

3 ★ Calculate the unknown angle. Y7 Ch 6 Challenge Q1

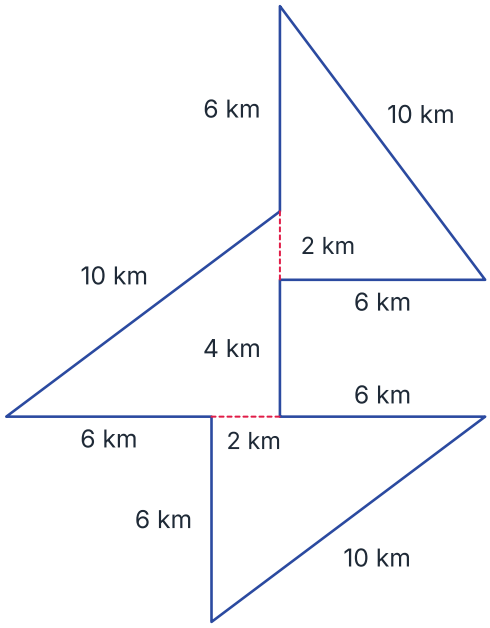


4 Calculate the perimeter and area of each of these figures, assuming that the non-slanting lines are either horizontal or vertical. Y7 Ch 9 Review Q8

a

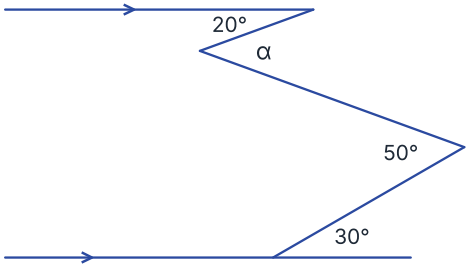


b

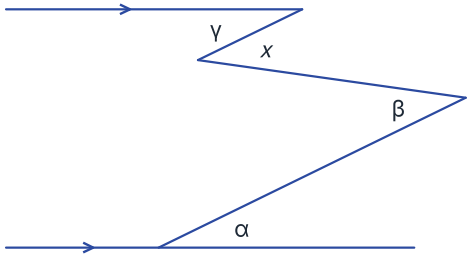


5 Y7 Ch 6 Challenge Q2

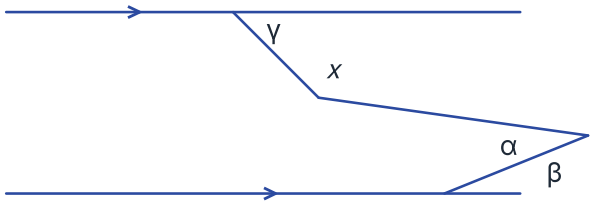
a Find the value of α.



b Find x in terms of α, β and γ.



6 Find x in terms of α, β and γ. Y7 Ch 6 Challenge Q3



7 A path 1 metre wide is being placed around the grassed area of a garden, as shown below. What is the area of the grassed part of the garden? What is its perimeter? What will the area of the path be? Y7 Ch 9 Challenge Q4

